

Questions

MathonGo

Q1 - 2024 (04 Apr Shift 1)

An effective power of a combination of 5 identical convex lenses which are kept in contact along the principal axis is $25D$. Focal length of each of the convex lens is :

- (1) 20 cm
- (2) 50 cm
- (3) 500 cm
- (4) 25 cm

Q2 - 2024 (04 Apr Shift 2)

A light ray is incident on a glass slab of thickness $4\sqrt{3}$ cm and refractive index $\sqrt{2}$. The angle of incidence is equal to the critical angle for the glass slab with air. The lateral displacement of ray after passing through glass slab is _____ cm. (Given $\sin 15^\circ = 0.25$)

Q3 - 2024 (05 Apr Shift 2)

Given below are two statements :

Statement I : When the white light passed through a prism, the red light bends lesser than yellow and violet.

Statement II : The refractive indices are different for different wavelengths in dispersive medium. In the light

of the above statements, chose the correct answer from the options given below;

- (1) Statement I is false but Statement II is true
- (2) Statement I is true but Statement II is false
- (3) Both Statement I and Statement II are true
- (4) Both Statement I and Statement II are false

Q4 - 2024 (06 Apr Shift 1)

The refractive index of prism is $\mu = \sqrt{3}$ and the ratio of the angle of minimum deviation to the angle of prism is one. The value of angle of prism is _____.

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Q5 - 2024 (06 Apr Shift 2)

In finding out refractive index of glass slab the following observations were made through travelling microscope 50 vernier scale division = 49MSD; 20 divisions on main scale in each cm

For mark on paper

$$\text{MSR} = 8.45 \text{ cm}, \text{VC} = 26$$

For mark on paper seen through slab

$$\text{MSR} = 7.12 \text{ cm}, \text{VC} = 41$$

For powder particle on the top surface of the glass slab

$$\text{MSR} = 4.05 \text{ cm}, \text{VC} = 1$$

(MSR = Main Scale Reading, VC = Vernier Coincidence)

Refractive index of the glass slab is :

(1) 1.52

(2) 1.35

(3) 1.42

(4) 1.24

Q6 - 2024 (06 Apr Shift 2)

For the thin convex lens, the radii of curvature are at 15 cm and 30 cm respectively. The focal length the lens is 20 cm. The refractive index of the material is :

(1) 1.2

(2) 1.8

(3) 1.5

(4) 1.4

Q7 - 2024 (08 Apr Shift 1)

Critical angle of incidence for a pair of optical media is 45° . The refractive indices of first and second media are in the ratio:

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(1) $1 : \sqrt{2}$

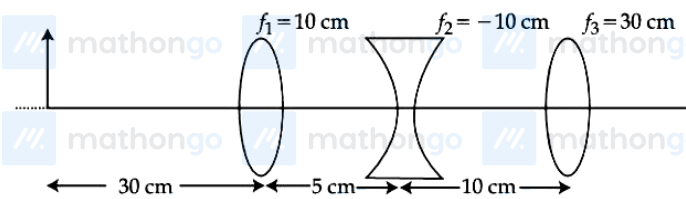
(2) $\sqrt{2} : 1$

(3) $2 : 1$

(4) $1 : 2$

Q8 - 2024 (08 Apr Shift 2)

The position of the image formed by the combination of lenses is :



(1) 15 cm (right of second lens)

(2) 30 cm (left of third lens)

(3) 15 cm (left of second lens)

(4) 30 cm (right of third lens)

Q9 - 2024 (09 Apr Shift 1)

Given below are two statements :

Statement (I) : When an object is placed at the centre of curvature of a concave lens, image is formed at the centre of curvature of the lens on the other side.

Statement (II) : Concave lens always forms a virtual and erect image.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

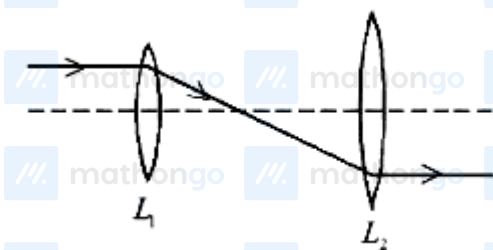
Q10 - 2024 (09 Apr Shift 2)

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Questions

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The following figure represents two biconvex lenses L_1 and L_2 having focal length 10 cm and 15 cm respectively. The distance between L_1 & L_2 is :



- (1) 10 cm
- (2) 35 cm
- (3) 25 cm
- (4) 15 cm

Questions

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Answer Key

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Q1 (1) mathongo Q2 (2) mathongo Q3 (3) mathongo Q4 (60) mathongo

Q5 (3) mathongo Q6 (3) mathongo Q7 (2) mathongo Q8 (4) mathongo

Q9 (3) mathongo Q10 (3) mathongo mathongo mathongo mathongo

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Solutions

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Q1

We know that $P_{eq} = \Sigma P_i$

$\therefore$ given all lenses are identical

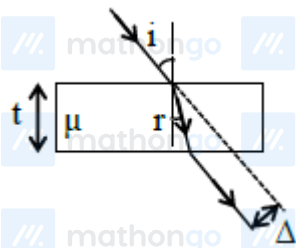
$$\therefore 5P = 25D$$

$$\therefore P = 5D$$

$$\therefore \frac{1}{f} = 5 \Rightarrow f = \frac{1}{5} \text{ m} = 20 \text{ cm}$$

Hence option (1)

Q2



$$i = \theta_c$$

$$\Rightarrow i = \sin^{-1}\left(\frac{1}{\mu}\right)$$

$$\Rightarrow i = 45^\circ$$

and according to snell's law

$$1 \sin 45^\circ = \sqrt{2} \sin r$$

$$\Rightarrow r = 30^\circ$$

$$\text{Lateral displacement } \Delta = \frac{t \sin(i-r)}{\cos r}$$

$$\Rightarrow \Delta = \frac{4\sqrt{3} \times \sin 15^\circ}{\cos 30^\circ}$$

$$\Rightarrow \Delta = 2 \text{ cm}$$

Q3

$$\text{As } \lambda_{\text{red}} > \lambda_{\text{yellow}} > \lambda_{\text{violet}}$$

Light ray with longer wavelength bends less.

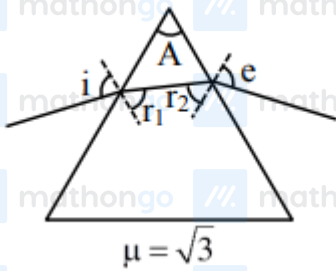
Q4

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Solutions

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For $\delta_{\min}$

$$i = e$$

$$r_1 = r_2 = \frac{A}{2}$$

$$\delta_{\min} = 1$$

$$\frac{A}{2i - A} = 1$$

$$2i = 2A$$

$$i = A$$

Snell's law

$$1 \times \sin i = \mu \sin r$$

$$\sin i = \mu \sin \left(\frac{A}{2} \right)$$

$$\sin A = \mu \sin \left(\frac{A}{2} \right)$$

$$2 \sin \frac{A}{2} \cos \frac{A}{2} = \sqrt{3} \sin \left(\frac{A}{2} \right)$$

$$\cos \left(\frac{A}{2} \right) = \frac{\sqrt{3}}{2}$$

$$\therefore \frac{A}{2} = 30^\circ$$

$$\therefore A = 60^\circ$$

Q5

$$1\text{MSD} = \frac{1 \text{ cm}}{20} = 0.05 \text{ cm}$$

$$1\text{VSD} = \frac{49}{50}\text{MSD} = \frac{49}{50} \times 0.05 \text{ cm} = 0.049 \text{ cm}$$

$$\text{LC} = 1\text{MSD} - 1\text{VSD} = 0.001 \text{ cm}$$

$$\text{For mark on paper, } L_1 = 8.45 \text{ cm} + 26 \times 0.001 \text{ cm} = 84.76 \text{ mm}$$

$$\text{For mark on paper through slab, } L_2 = 7.12 \text{ cm} + 41 \times 0.001 \text{ cm} = 71.61 \text{ mm}$$

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Solutions

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For powder particle on top surface, $ZE = 4.05 \text{ cm} + 1 \times 0.001 \text{ cm} = 40.51 \text{ mm}$

$$\therefore \text{actual } L_1 = 84.76 - 40.51 = 44.25 \text{ mm}$$

$$\text{actual } L_2 = 71.61 - 40.51 = 31.10 \text{ mm}$$

$$L_2 = \frac{L_1}{\mu}$$

$$\Rightarrow \mu = \frac{L_1}{L_2} = \frac{44.25}{31.10} = 1.42$$

Q6

$$\frac{1}{f} = \left(\frac{\mu_{\text{lens}}}{\mu_{\text{air}}} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{+20} = \left(\frac{\mu}{1} - 1 \right) \left(\frac{1}{+15} - \frac{1}{(-30)} \right)$$

$$\Rightarrow \frac{1}{20} = (\mu - 1) \left(\frac{3}{30} \right)$$

$$\Rightarrow \mu - 1 = \frac{1}{2}$$

$$\Rightarrow \mu = 1 + \frac{1}{2} = \frac{3}{2} = 1.5$$

Q7

$$\sin \theta_c = \frac{\mu_R}{\mu_d} = \frac{\mu_2}{\mu_1}$$

$$\sin 45^\circ = \frac{\mu_2}{\mu_1}$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{\mu_2}{\mu_1}$$

$$\Rightarrow \frac{\mu_1}{\mu_2} = \frac{\sqrt{2}}{1}$$

Q8

For lens 1 : $f_1 = 10$, $u = -30$, $v = ?$

$$v = \frac{uf}{u+f} = \frac{-30 \times 10}{-30+10} = 15$$

For lens 2: $f_1 = -10$, $u = 10$, $v = ?$

$$v = \frac{uf}{u+f} = \frac{10 \times -10}{10-10} = \infty$$

For lens 3 : $f = 30$, $u = -\infty$, $v = ?$

So v will be 30 .

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Solutions

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Q9

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

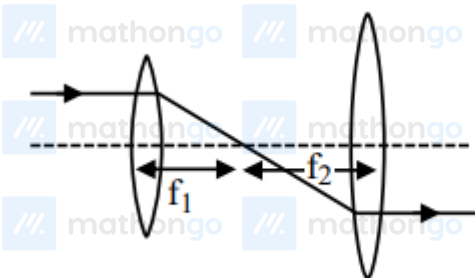
$$\frac{1}{v} - \frac{1}{-2f} = \frac{1}{-f}$$

$$\Rightarrow \frac{1}{v} = \frac{-1}{2f} \Rightarrow v = -2f$$

$$\frac{1}{v} = \frac{1}{u} + \frac{1}{f} \Rightarrow \text{Virtual image of Real object.}$$

In statement II, it is not mentioned that object is real or virtual hence Statement II is false.

Q10



$$D = f_1 + f_2 = 25 \text{ cm}$$

Paraxial parallel rays pass through focus and ray from focus of convex lens will become parallel

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